

Ayudantía 16/05

1- $\vec{F} = x\hat{i} + y\hat{j}$; $x = 5t^2 - 1$, $y = 3t^3 + 2$, $m = 3.00 \text{ kg}$

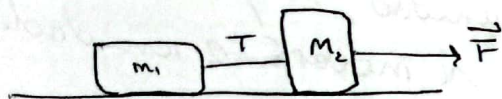
$$\Rightarrow \sum \vec{F} = m\vec{a} = m \frac{d^2 \vec{r}}{dt^2} = m \frac{d^2}{dt^2} ((5t^2 - 1)\hat{i} + (3t^3 + 2)\hat{j})$$

$$= m \frac{d}{dt} (10t\hat{i} + 9t^2\hat{j}) = m(10\hat{i} + 18t\hat{j})$$

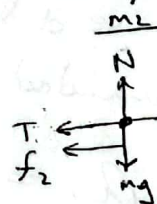
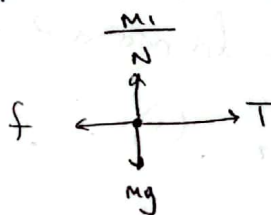
$$t = 2.00 \text{ s} \Rightarrow \vec{F} = (3.00 \text{ kg})(10 \text{ m/s}^2 \hat{i} + 36 \text{ m/s}^2 \hat{j})$$

$$= (30.0\hat{i} + 108\hat{j}) [\text{N}]$$

2-



1) Diagrama cuerpo libre $\mu_s = 0.2$



$\mu_k = 0.1$
 $a = 2.0 \text{ m/s}^2$

2) Ecuaciones

$$\underline{m_1} \quad \sum F_x = 0$$

$$T - f_1 = 0 \quad (1)$$

$$\underline{m_2}$$

$$\sum F_x = 0 \Rightarrow F - T - f_2 = 0 \quad (3)$$

$$\sum F_y = 0 \Rightarrow N_1 - m_1 g = 0 \quad (2)$$

$$\sum F_y = 0 \Rightarrow N_2 - m_2 g = 0 \quad (4)$$

$$F_{rota} = \mu N$$

$$f_1 = \mu N_1, \quad f_2 = \mu N_2$$

Fuerza máxima: (1) $T = f_1 = \mu N_1 = \mu_s m_1 g$

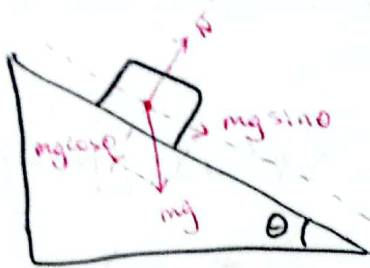
$$(3) F = T + f_2 = \mu_s m_1 g + \mu_s m_2 g = 31.36 \text{ N}$$

Si $a = 2.0 \text{ m/s}^2$, T y F

$$\sum F_x = m_1 a \Rightarrow T - f_1 = m_1 a \Rightarrow T = m_1 a + f_2 = m_1 a + \mu_k m_1 g = 17.88 \text{ N}$$

$$\sum F_x = m_2 a \Rightarrow F - T - f_2 = m_2 a \Rightarrow F = m_2 a + T + f_2 = 47.68 \text{ N}$$

3-



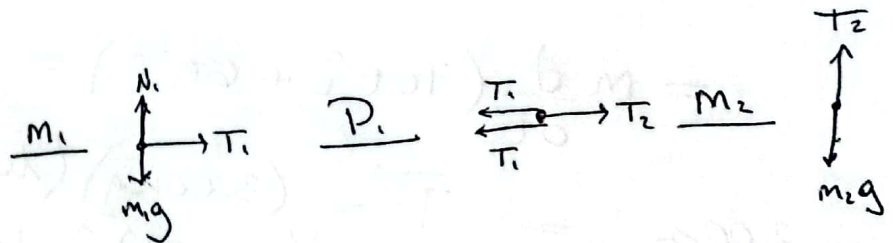
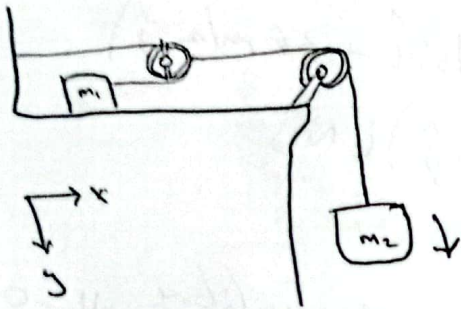
$$\Sigma F_x = ma_x \quad ; \quad x = 2.0 \text{ m}$$

$$mg \sin \theta = ma \Rightarrow a = g \sin \theta$$

$$a) \quad a = 2.5 \text{ m/s}^2$$

$$b) \quad v_f^2 = v_o^2 + 2ax \Rightarrow v_f = \sqrt{v_o^2 + 2ax} = \sqrt{2ax} = 3.16 \text{ m/s}$$

4-



a) Si m_1 se mueve a la derecha en 1 unidad, la pda 2 también se moverá 1 unidad. $\therefore$ la masa 2 se moverá 2 unidades $\therefore a_1 = 2a_2$ (1)

b) m_1

$$\Sigma F_x = m_1 a_1$$

$$T_1 = m_1 a_1 \quad (2)$$

 P_1

$$\Sigma F_x = m_p a = 0$$

$$T_2 - 2T_1 = 0 \Rightarrow T_2 = 2T_1 \quad (3)$$

 m_2

$$\Sigma F_y = m_2 a_2$$

$$m_2 g - T_2 = m_2 a_2$$

$$T_2 = m_2 g - m_2 a_2 \quad (4)$$

 $T_1, T_2?$

$$(1) \rightarrow (4) \quad T_2 = m_2 g - \frac{m_2 a_1}{2} \quad ; \quad (2) \quad a_1 = \frac{T_1}{m_1}$$

$$\Rightarrow T_2 = m_2 g - \frac{m_2 T_1}{2 m_1} \quad ; \quad (3) \quad T_1 = \frac{T_2}{2} \Rightarrow T_2 = m_2 g - \frac{m_2 T_2}{4 m_1}$$

$$\Rightarrow 4 m_1 T_2 = 4 m_1 m_2 g - m_2 T_2 \Rightarrow T_2 (4 m_1 + m_2) = 4 m_1 m_2 g$$

$$\Rightarrow T_2 = \frac{4 m_1 m_2 g}{4 m_1 + m_2}$$

$$\Rightarrow T_1 = \frac{2 m_1 m_2 g}{4 m_1 + m_2}$$

$$a_1 = \frac{2 m_2 g}{4 m_1 + m_2}$$

$$a_2 = \frac{m_2 g}{4 m_1 + m_2}$$